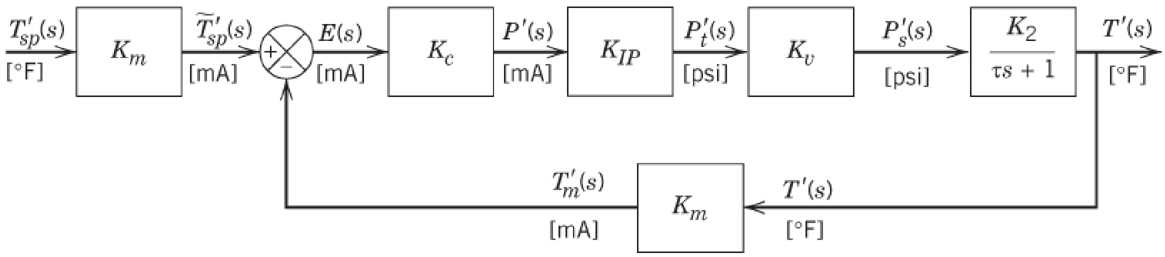
Homework Set 5. *Feedback Loop, Closed-Loop Stability and Performance*

(Due on Monday, March 17, 2025)

1. (20 pt) Closed-loop transfer functions. [Seborg 11.3]

Consider proportional-only control of the stirred-tank heater control system in Fig. E11.13. The temperature transmitter is $K_m = 4$ mA/50 °F. The controller has a gain of 5, while the gains for the control valve and current-to-pressure transducer are $K_v = 1.2$ (dimensionless) and $K_{IP} = 0.75$ psi/mA, respectively. The time constant for the tank is $\tau = 5$ min. After the set point is changed from 80 to 85 °F, the tank temperature eventually reaches a new steady-state value of 84.14 °F from 80 °F, which was measured with a highly accurate thermometer.



(a) What is the offset (i.e., $\lim_{t\to\infty}\left[y(t) - y_{sp}(t)\right]$ under a unit step change in $y_{sp}(t)$)?

(b) What is the process gain K_2?

(c) What is the pressure signal p_t' (in deviation form) to the control valve at the final steady state?

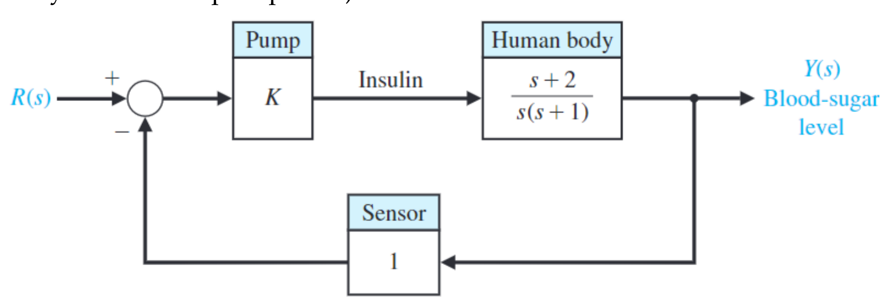
2. (20 pt) Simple proportional controller design. [Dorf E5.7]

Effective control of insulin injections can result in better lives for diabetic persons. Automatically controlled insulin injection by means of a pump and a sensor that measures blood sugar can be very effective. A pump and injection system has a feedback control as shown in the figure below. Here $R(s)$ is the desired blood-sugar level (namely $Y_{sp}(s)$) and $Y(s)$ is the actual blood-sugar level.

(a) Calculate the closed-loop transfer function $Y(s)/R(s)$.

(b) What is the offset?

(c) What is the maximum K such that the response does not overshoot? (Hint: If analytical solution appears to be difficult, you may simulate the response under a range of K and calculate overshoot automatically from the step response.).



3. (20 pt) Closed-loop poles and closed-loop stability by characteristic equation. [Dorf P6.11]

A feedback control system has a characteristic equation

$$s^3 + (1 + K)\, s^2 + 3s + (1 + 7K) = 0.$$

The parameter K must be positive.

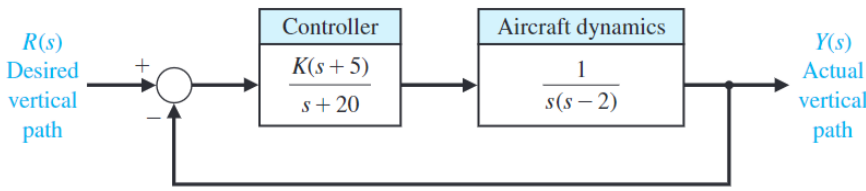
(a) What is the maximum value K can assume before the system becomes unstable?

(b) When K is equal to the maximum value, the system oscillates. Determine the frequency of oscillation.

(c) When K is 50% of the maximum value, determine the roots of the characteristic equation (namely the *closed-loop poles* – poles of the closed-loop transfer functions).

4. (20 pt) Stabilization of an unstable process. [Dorf P6.19]

The goal of *vertical takeoff and landing (VTOL)* aircraft is to achieve operation from relatively small airports and yet operate as a normal aircraft in level flight. An aircraft taking off in a form similar to a rocket is inherently unstable. A control system using adjustable jets can control the vehicle, as shown in the figure below.

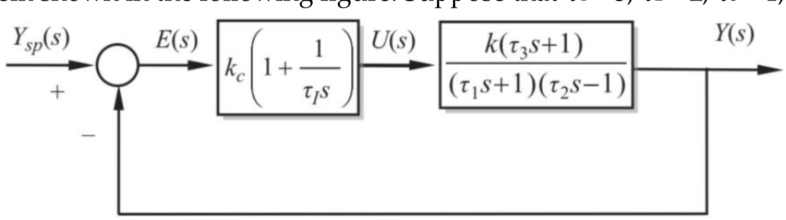


(a) Determine the range of controller gain for which the feedback control system is stable.

(b) Determine the gain K for which the feedback control system is marginally stable, as well as the roots of the characteristic equation for this value.

5. (20 pt) Controller tuning based on simulated performance. [Kravaris 16.10]

Consider the system shown in the following figure. Suppose that $\tau_1 = 3$, $\tau_2 = 2$, $\tau_3 = 1$, and $k = 1$.



(a) For $\tau_I = 3$ and *various* values of k_c, simulate the response of $y(t)$ and $u(t)$ to a unit step in $y_{sp}(t)$, and calculate the performance indices (ISE and ISC). Plot (ISE+ISC) as a function of controller gain k_c.

(b) Vary τ_I and repeat part (a). Place all the curves along with the result of (a) on a single plot.

(c) Determine the best PI controller with respect to the performance ISE+ISC.

6. (Bonus: 10 pt) Process control in practice: Kicks in PID controllers and PID in *digital versions*. [Seborg 8.8]

When there is a sudden change in the setpoint, the derivative mode in PID controllers may cause a "kick" in the controller output $p(t)$. To see how this happens and how it can be desirably eliminated in practice, please read Section 8.3 of Seborg. Also, since controllers in the real world never receives feedback in continuous time, actual "digital controllers" must be considered in discretized "sampling times". Please read Section 8.6 of Seborg. Then, answer the following questions.

(a) Find an expression for the amount of derivative kick that will be applied to the process for the position form of the PID digital algorithm (Eq. 8-25) if a set-point change of magnitude Δy_{sp} is made between the $k-1$ and k sampling instants.

(b) Repeat for the proportional kick, that is, the sudden change caused by the proportional mode.

(c) Sketch the sequence of controller output values at the $k-1$, k,… sampling times for the case of a set-point change of Δy_{sp} magnitude made just after the $k-1$ sampling time if the controller receives a constant measurement $\bar{y}_m$ and the initial set point is $\bar{y}_{sp} = \bar{y}_m$. Assume that the controller output initially is $\bar{p}$.

(d) How can Eq. 8-25 be modified to eliminate derivative kick?

7. (Bonus: 10 pt) Process control in depth: Tuning controller parameters by *rigorous optimization* of controller performance. *You may or may not need to use Matlab.* [Kravaris 16.2]

Consider a process to be controlled with a PI controller whose parameters are to be determined:

$$G_p(s) = \frac{-s+4}{s^2+3s+2},\ G_c(s) = K_c\left(1+\frac{1}{\tau_I s}\right),\ G_v = G_m = 1.$$

(a) Calculate closed-loop transfer functions $Y(s)/Y_{sp}(s)$ and $U(s)/Y_{sp}(s)$. Determine the necessary and sufficient conditions for closed-loop stability, using Routh array.

(b) Newton, Gould, and Kaiser (1957) provided the formulas to relate ISE to $Y(s)/Y_{sp}(s)$. See the table on the next page. For example, if $Y(s)/Y_{sp}(s) = (b_0 s + b_1)/(s + a_1)$, then ISE = $(1/2)(\varphi_0^2/a_1)$ where $\varphi_0 = b_1/a_1 - b_0$. The same formulas can be used to relate ISC to $U(s)/Y_{sp}(s)$. For the system given, please determine the expressions of ISE and ISC in terms of K_c and τ_I.

(c) Determine the two controller parameters by minimizing ISE + ISC (as functions of the two parameters as determined in part (b)), subject to the stability constraints (as determined in part (a)). You may need to know Matlab command `fmincon`.

System order	Closed-loop transfer function
1	$\dfrac{b_0 s + b_1}{s + a_1}$
	$J = \dfrac{1}{2}\dfrac{\varphi_0^2}{a_1}$ *where* $\varphi_0 = \dfrac{b_1}{a_1} - b_0$
2	$\dfrac{b_0 s^2 + b_1 s + b_2}{s^2 + a_1 s + a_2}$
	$J = \dfrac{1}{2}\dfrac{\varphi_0^2 a_2 + \varphi_1^2}{a_1 a_2}$ *where* $\varphi_0 = \dfrac{b_2}{a_2} - b_0,\ \varphi_1 = \dfrac{b_2}{a_2} a_1 - b_1$
3	$\dfrac{b_0 s^3 + b_1 s^2 + b_2 s + b_3}{s^3 + a_1 s^2 + a_2 s + a_3}$
	$J = \dfrac{1}{2}\dfrac{\varphi_0^2 a_2 a_3 + (\varphi_1^2 - 2\varphi_0\varphi_2) a_3 + \varphi_2^2 a_1}{(a_1 a_2 - a_3) a_3}$ *where* $\varphi_0 = \dfrac{b_3}{a_3} - b_0,\ \varphi_1 = \dfrac{b_3}{a_3} a_1 - b_1,\ \varphi_2 = \dfrac{b_3}{a_3} a_2 - b_2$
4	$\dfrac{b_0 s^4 + b_1 s^3 + b_2 s^2 + b_3 s + b_4}{s^4 + a_1 s^3 + a_2 s^2 + a_3 s + a_4}$
	$J = \dfrac{1}{2}\dfrac{\varphi_0^2 \left(a_2 a_3 - a_1 a_4\right) a_4 + \left(\varphi_1^2 - 2\varphi_0\varphi_2\right) a_3 a_4 + \left(\varphi_2^2 - 2\varphi_1\varphi_3\right) a_1 a_4 + \varphi_3^2 \left(a_1 a_2 - a_3\right)}{\left(a_1 a_2 a_3 - a_3^2 - a_1^2 a_4\right) a_4}$ *where* $\varphi_0 = \dfrac{b_4}{a_4} - b_0,\ \varphi_1 = \dfrac{b_4}{a_4} a_1 - b_1,\ \varphi_2 = \dfrac{b_4}{a_4} a_2 - b_2,\ \varphi_3 = \dfrac{b_4}{a_4} a_3 - b_3$